Guarded Drift-Aware Background Tracking for Synthetic Radiation-Portal Count Streams: A Reproducible Benchmark and a Negative Result

Rad-DAES Project

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Abstract

Radiation-portal monitors must separate source-like count excursions from a drifting background and from telemetry faults. This draft evaluates *Rad-DAES*, a candidate pipeline that combines a four-condition telemetry-integrity gate, a pre-update Mahalanobis detector, and the scalar drift-aware step size of the supplied DAES reference implementation. The study is synthetic-only and deliberately narrow: six-window Poisson streams under four background regimes and eight telemetry-fault conditions, tuned on development seeds only, with the recurring regime held out, and evaluated on 32 scenarios with five seeds and nine methods (1,440 runs). The result is negative. No deployable method achieved both useful event detection and an acceptable false-alarm rate. The best mean sample-level F1 is 0.242 for the non-deployable oracle gate; Rad-DAES obtains 0.023 with 18.508 false alarms per hour, and the four methods whose development configurations met the false-alarm constraint obtain F1 at most 0.003 and detect at most 6% of source events. The traces show why: after an abrupt background step the innovation guard freezes the model for hundreds of seconds, the 90-sample sustained-release rule almost never fires for Rad-DAES (below 0.01% of its samples), and the adaptive background absorbs source energy during the ramp. These are simulation results from an unvalidated generator. They do not establish field readiness, operational safety, or novelty, and no statistical-significance claim is made.

1 Introduction

A radiation portal monitor observes a stream of gamma counts in several energy windows. A source that passes through the portal raises some windows for a short interval; background radon, weather, cargo, and detector drift change the same windows over minutes or hours; and a telemetry fault can make an observation invalid, late, stuck, or implausibly dispersed. A background tracker that adapts too eagerly can absorb the source it is meant to detect. A tracker that adapts too slowly leaves a drift-induced residual that looks like a source. The engineering problem is therefore a coupling between detection, adaptation, and data quality, not a standalone optimizer problem.

This draft asks a narrow question: under matched synthetic streams, does a guarded DAES pipeline improve the detection/false-alarm/stability trade-off relative to static, EWMA, gated EWMA, RLS, Adam, ungated DAES, fixed-step gated DAES, and an oracle gate? The answer reported here is no, and the more useful output is a reproducible account of why. The DAES component is the original scalar step-size heuristic supplied with the project brief. The matrix update that appears in some generated planning material is explicitly not implemented as DAES; the discrepancy is documented in the repository.

The paper makes four bounded contributions:

1. a reproducible synthetic benchmark with labelled background regimes, source events, and telemetry faults, including a held-out recurring regime;
2. a documented separation between the original DAES scalar controller, the Gaussian quasi-likelihood curvature used by the wrapper, and the telemetry gate;
3. a complete negative-and-mixed report: all methods and scenarios are reported, including the cases where the candidate loses;
4. a mechanistic account, supported by saved per-step traces, of two failure modes: gate-induced freezing under drift and adaptive-background contamination during source ramps.

No real radiation data are used. The data-access blockers are recorded in the repository. The experiments are a first draft, not a field validation.

2 Related work

Background estimation and source detection. Radiation-portal analysis has moved toward large, realistic datasets and reproducible algorithm development [5, 6]. Detection methods range from moving-source algorithms over portal networks [3] to studies of cargo-induced background suppression [14] and standardised portal data formats [1]. These works motivate the six-window count representation and the concern with background drift, but they do not provide a local stream on which the present pipeline can be validated. The synthetic generator is therefore an explicit placeholder.

Drift, smoothing, and change detection. Concept-drift adaptation distinguishes transient noise, gradual change, abrupt change, and recurring change [4]. The EWMA control chart [13] and the CUSUM procedure [11] are classical tools for the same problem. The gate in this draft uses a sliding-window dispersion check and an innovation threshold rather than a full CUSUM, but the design tension is the same: a persistent shift should eventually be tracked, while a short source-like excursion should not contaminate the background.

Step sizes and robust estimation. Polyak-style gradient methods and two-point step-size rules provide the optimisation background for adaptive step sizes [12, 2]. The supplied DAES implementation is a scalar heuristic in this family: it estimates the largest curvature eigenvalue with power iteration and scales the step by its inverse. Robust location estimation [7] and model-based fault diagnosis [8] motivate the separation between an outlier-resistant update and a fault detector. Adam [9] and recursive least squares [10] are included as optimisation/estimation baselines, not as claims about radiation physics.

3 Methods

3.1 Synthetic streams

Each stream is a sequence of $D = 6$ non-negative count vectors at 1 Hz. The six energy windows and nominal count rates are design choices taken from the project brief, not calibrated detector properties. For window j and time t,

$$x_{t,j} \sim \mathrm{Poisson}\big(\mu^{\mathrm{true}}_{t,j} + s_{t,j}\big), \tag{1}$$

where μ_t^{true} is a labelled background mean and s_t is a labelled source excess. Poisson draws are made after the background and source terms are combined. Telemetry faults are applied afterwards.

The background regimes are:

- `stable`: a small AR(1) fluctuation around the nominal rate;
- `gradual`: a slow sinusoidal trend plus a linear trend;
- `abrupt`: two multiplicative step changes with different per-window factors;
- `recurring`: three broad bumps, strongest in the low- and high-energy windows.

The recurring regime is held out from tuning. Each 6 h stream attempts six source events of 20–60 s, with amplitudes 3–8 baseline standard deviations and three templates. The excess follows a $\sin^2$ profile, so a labelled event has low amplitude at both ends and a narrow peak. Fault conditions are `clean`, `mixed_5`, `mixed_10`, `mixed_15`, and single-mode `dropout_10`, `jitter_10`, `stuck_10`, and `spike_10`; the suffix is the approximate fraction of affected samples. The generator, labels, and seeds are deterministic.

3.2 Background model and detector

The background state is a mean μ_t and covariance Σ_t. Detection uses the pre-update innovation

$$r_t = x_t - \mu_{t-1}, \qquad s_t = r_t^\top \big(\Sigma_{t-1} + \epsilon I\big)^{-1} r_t, \tag{2}$$

with $\epsilon = 10^{-6}$. An alarm requires two consecutive scores above a calibrated threshold τ_{alarm}. The detector never sees the updated background for the current sample: detection precedes adaptation. The detector is evaluated on every sample, including samples that the telemetry gate rejects for updating; a high score can therefore alarm and freeze the model at the same time.

The Gaussian quasi-likelihood

$$L_t(\theta) = \tfrac{1}{2}(x_t - \theta)^\top \Sigma_{t-1}^{-1}(x_t - \theta) \tag{3}$$

has gradient $-\Sigma_{t-1}^{-1} r_t$ and Hessian $H_t = \Sigma_{t-1}^{-1}$. The covariance is tracked separately. On an eligible sample, with post-update residual $u_t = x_t - \mu_t$,

$$\tilde{\Sigma}_t = (1-\beta)\Sigma_{t-1} + \beta u_t u_t^\top, \tag{4}$$

$$\Sigma_t = (1-\rho)\tilde{\Sigma}_t + \rho\, \mathrm{diag}(\tilde{\Sigma}_t), \tag{5}$$

$$\mathrm{diag}(\Sigma_t) \leftarrow \max\big(\mathrm{diag}(\Sigma_t),\, \max(10^{-6}, 0.25\,\mu_0)\big), \tag{6}$$

with $\beta = 0.02$, $\rho = 0.10$, and μ_0 the nominal count vector. The diagonal floor is a regularisation choice; it is not a claim that the true covariance is Poisson. The covariance time constant is 50 samples, comparable to the 20–60 s source duration, which is relevant to the results below.

3.3 Telemetry-integrity gate

The gate is a conjunction of four conditions evaluated on the raw observation and a length-10 sliding buffer:

1. **validity**: every channel is finite and strictly positive;
2. **timing**: $0.5 \le \Delta t_t \le 1.5\,\text{s}$;

3. **Poisson dispersion**: $0.25 \leq \mathrm{Var}_W(x_j)/\mathrm{Mean}_W(x_j) \leq 3.5$ in every window;

4. **innovation**: $s_t \leq \tau_{\mathrm{gate}}$, or a sustained-exceedance release is active.

The release activates after 90 consecutive samples that pass validity, timing, and dispersion but exceed τ_{gate}, and resets on any other telemetry failure or on a score at or below threshold. The patience is longer than the longest synthetic source event (60 s). It is a design choice added to prevent a persistent background step from freezing the mean and producing an unbounded false-alarm burst. The gate is separate from the detector: a sample can alarm and still be rejected for updating, and the evaluation reports both decisions.

3.4 Original DAES controller

The supplied DAES implementation is a scalar step-size heuristic for gradient descent on a drifting strongly convex quadratic. It estimates $\lambda_{\max}(H_t)$ by k power iterations, computes the Rayleigh quotient r_t, and uses

$$d_t = \frac{|r_t - r_{t-1}|}{\max(|r_{t-1}|, 10^{-12})}, \tag{7}$$

$$\alpha_t = \mathrm{clip}(\alpha_0 + \kappa d_t, \alpha_0, \alpha_{\max}), \tag{8}$$

$$\hat{\lambda}_t = (1 - \alpha_t)\hat{\lambda}_{t-1} + \alpha_t r_t, \tag{9}$$

$$\eta_t = \frac{\gamma}{(1+s)\hat{\lambda}_t}, \tag{10}$$

with an upward correction $\hat{\lambda}_t \leftarrow \max(\hat{\lambda}_t, r_t)$ when $r_t > \hat{\lambda}_{t-1}$. The reference defaults are $\gamma = 1.78$, $s = 0.10$ (a fixed safety margin, distinct from the score s_t in Eq. (2)), $k = 7$, $\alpha_0 = 0.16$, $\alpha_{\max} = 0.82$, and $\kappa = 7.0$. Rad-DAES tunes γ on development data; the equation form is unchanged.

3.5 Rad-DAES wrapper and baselines

When the gate passes, Rad-DAES applies

$$\mu_t = \mu_{t-1} + \eta_t \Sigma_{t-1}^{-1} r_t, \tag{11}$$

which is gradient descent on L_t with the scalar DAES step. When the gate rejects, the mean and covariance are frozen. The baselines are:

- `static`: initial mean and covariance frozen;
- `ewma`: fixed-α EWMA mean and the same covariance update;
- `gated_ewma`: EWMA behind the gate;
- `rls`: recursive least squares with forgetting;
- `adam`: Adam on the Gaussian quasi-likelihood gradient;
- `daes`: ungated DAES;
- `gated_daes_fixed`: gate plus a DAES step frozen at initialisation;
- `oracle_gate`: gate replaced by the true source/fault labels; non-deployable reference.

All methods share the detector, windows, covariance rule where applicable, and calibration procedure.

Table 1: Held-out results. F1 is the mean of 32 scenario means; the interval is a two-sided 95% Student-t interval across scenario means. FAR is false alarms per hour over non-source time. BG is mean relative background error. Recovery is the mean recovery time after abrupt steps (— when no event recovered). Runtime is wall-clock milliseconds per sample, excluding file I/O.

Method	F1 (95% CI)	FAR/h	BG rel.	Recall	Precision	Recovery (s)	ms/sample
Static	0.148 [0.109, 0.188]	472.4	0.0546	0.370	0.249	—	0.139
EWMA	0.003 [0.002, 0.004]	2.48	0.0153	0.002	0.089	399.1	0.267
Gated EWMA	0.095 [0.082, 0.108]	102.7	0.0174	0.121	0.473	1734	0.247
RLS	0.003 [0.002, 0.004]	2.48	0.0153	0.002	0.089	399.1	0.271
Adam	0.003 [0.002, 0.004]	2.50	0.0278	0.002	0.078	5286	0.298
DAES	0.002 [0.001, 0.003]	2.03	0.0375	0.001	0.054	4564	0.442
Gated DAES (fixed)	0.006 [0.004, 0.008]	6.65	0.0689	0.004	0.054	5196	0.300
Rad-DAES	0.023 [0.018, 0.028]	18.51	0.0254	0.024	0.142	4198	0.359
Oracle gate	0.242 [0.230, 0.254]	7.29	0.0143	0.158	0.721	4213	0.368

3.6 Protocol and estimands

The held-out matrix is 4 regimes × 8 fault conditions × 5 seeds = 160 streams; each stream is processed by nine methods, giving 1,440 runs. Development tuning uses five seeds, three regimes (`stable`, `gradual`, `abrupt`), and `mixed_10` faults. The tuning grid contains 69 candidate configurations. For each method, the selected configuration maximises mean development F1 subject to mean background relative error ≤ 0.10 and mean false alarms per hour ≤ 2.0; if no candidate is feasible, the least-violating candidate is used and the fallback is reported. Four methods had a feasible candidate, four used the fallback rule, and `static` has no tunable parameter and is marked fixed.

Alarm thresholds are the empirical $1 - 1/3600$ quantile of pre-update scores on clean stable development streams, pooled over five seeds; the gate threshold is the 0.999 quantile with a floor of 10. Both are frozen before test. The primary estimand is mean sample-level F1 across the 32 test scenarios, each weighted equally. Secondary estimands are false alarms per hour over non-source time, background relative error on non-source/non-fault samples, event-level detection (the fraction of the 960 source events with at least one alarm), threshold-free ROC and PR AUC computed from the pre-update score, recovery time after abrupt steps, detection latency, update stability, runtime, and gate rejection rates. Intervals are two-sided 95% Student-t intervals across seeds or scenarios as indicated; no multiplicity correction is applied and no significance claim is made.

4 Results

Table 1 reports the primary estimand and the main secondary outcomes. Rad-DAES obtains mean F1 0.023 (95% CI $[0.018, 0.028]$), mean false alarms 18.508/h, and mean background relative error 0.025. The best observed method is Oracle gate with mean F1 0.242, but that method uses the true source and fault labels and is not deployable. Among deployable methods, `static` has the highest F1 (0.148) at 472.4/h, which is more than two orders of magnitude above the development target. The paired differences in Table 3 are descriptive; they are not corrected for multiple comparisons.

The operating point dominates the ranking. The four methods whose development configurations satisfied the false-alarm constraint (EWMA, RLS, Adam, ungated DAES) have test F1 at most 0.003 and detect at most 6% of source events, with test FAR between 2.0 and 2.5/h. They are,

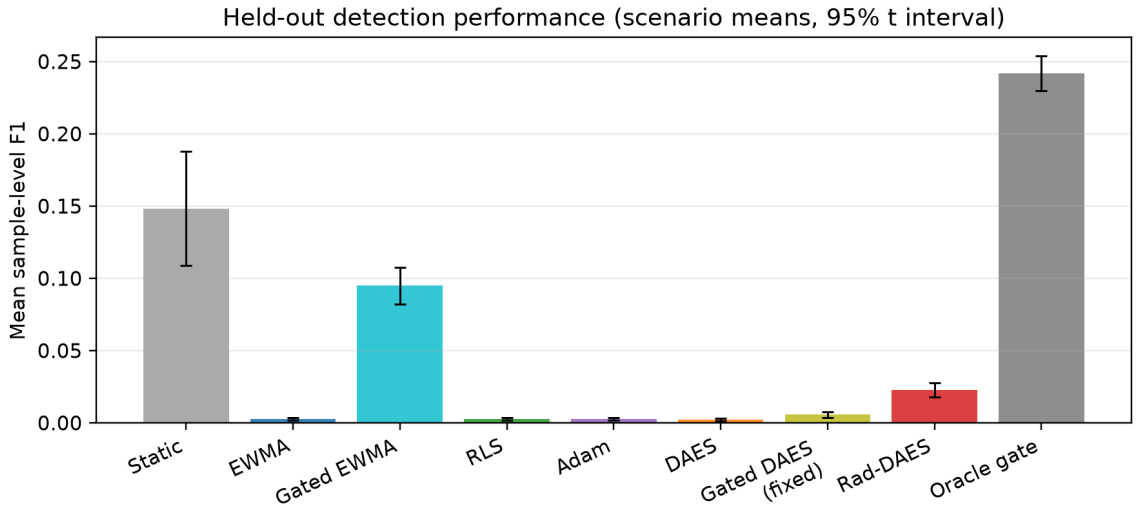


Figure 1: Mean sample-level F1 by method. Error bars are 95% Student-t intervals across the 32 held-out scenario means. The oracle gate is a non-deployable reference.

Table 2: Threshold-free discrimination and gate behaviour. ROC/PR AUC use the pre-update score and do not depend on the alarm threshold. Event detection is the fraction of 960 source events with at least one alarm. Alarm fraction is the fraction of all evaluated samples that alarm. Release rate is the fraction of samples in which the sustained-exceedance release is active.

Method	ROC AUC	PR AUC	Event det.	Alarm frac.	Gate rej. clean	Gate rej. source	Release rate
Static	0.728	0.128	0.821	0.1341	0.397	0.630	0.0017
EWMA	0.622	0.031	0.060	0.0007	0.260	0.333	0.0000
Gated EWMA	0.691	0.089	0.406	0.0296	0.275	0.406	0.0000
RLS	0.622	0.031	0.060	0.0007	0.260	0.333	0.0000
Adam	0.637	0.033	0.056	0.0007	0.260	0.333	0.0000
DAES	0.600	0.024	0.054	0.0006	0.260	0.330	0.0000
Gated DAES (fixed)	0.558	0.019	0.059	0.0019	0.260	0.326	0.0000
Rad-DAES	0.629	0.036	0.105	0.0054	0.261	0.338	0.0000
Oracle gate	0.773	0.223	0.736	0.0039	0.260	0.485	0.0000

in effect, blind at the calibrated operating point. The methods with visible detection either violate the false-alarm constraint or use labels: gated EWMA reaches F1 0.095 at 102.7/h, and Rad-DAES reaches 0.023 at 18.508/h. `static` reaches 0.148 only by alarming on 13.4% of all samples. There is no deployable method in the matrix that is both sensitive and quiet.

Table 2 separates threshold choice from separability. The oracle gate has the highest ROC AUC (0.773) and PR AUC (0.223), followed by `static` (0.728 and 0.128) and gated EWMA (0.691 and 0.089). Rad-DAES obtains ROC AUC 0.629 and PR AUC 0.036, below both gated EWMA and `static`. The ordering is therefore not an artefact of one threshold: the DAES-based methods rank source samples less well than the simpler alternatives on these streams.

Regime and fault structure. Figures 2 and 3 show that performance is not uniform. The recurring regime is the only regime not used for tuning; it is therefore the clearest check of whether a configuration generalises to a different drift shape. It does not rescue the candidate: Rad-DAES remains below gated EWMA and far below the oracle. Fault conditions change which samples are

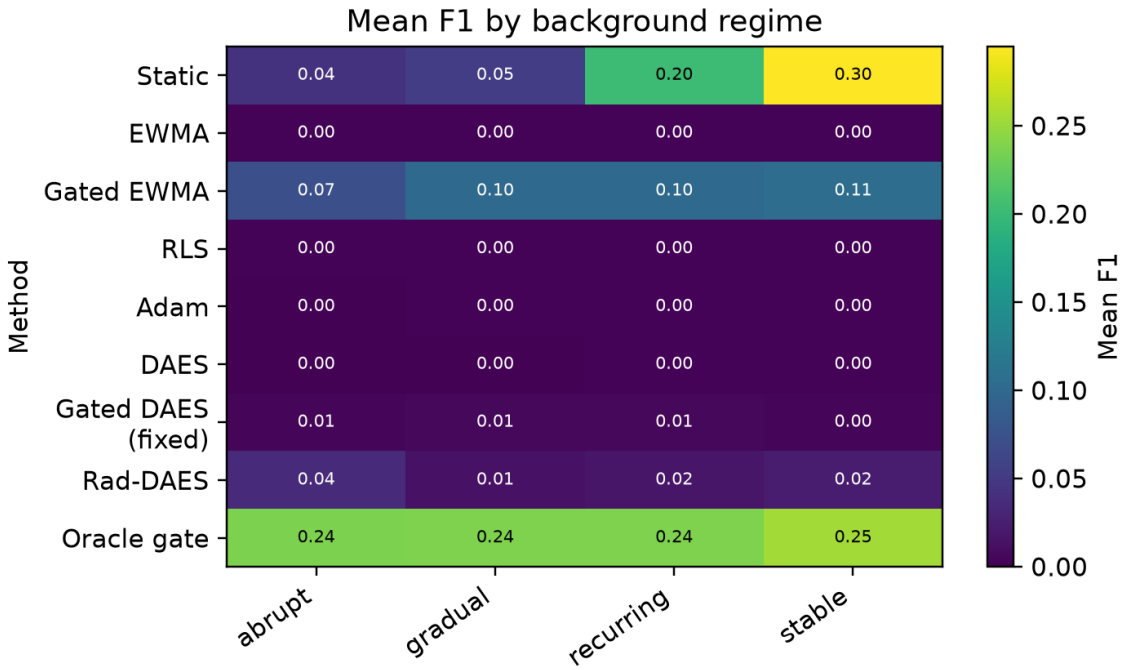


Figure 2: Mean F1 by background regime. Each cell is the mean over the eight fault conditions and five seeds.

eligible for updating, so a method that looks adequate on clean streams can lose when dropout, jitter, stuck, or spike faults are present. The spike condition is the worst for false alarms: Rad-DAES reaches 18.508/h on average but 68.8/h under `spike_10`, because a spike both alarms and freezes the update.

Gate behaviour. The gate rejects 26.1% of clean samples and 33.8% of source samples for Rad-DAES, and the sustained release is active on about 3×10^{-6} of samples. The release is not merely rare: in the representative trace (Figure 6) the consecutive-exceedance counter reaches 89 samples and then resets, one sample short of the 90-sample patience, so the model stays frozen through the step. The tracked mean remains near its pre-step value for roughly 350 s while the true background has already moved by about 170 counts/s. The gate's source-rejection rate is the expected cost of protecting the background, but here the same mechanism also protects the background from the drift it is supposed to follow.

Recovery and stability. Figure 7 reports recovery time after abrupt steps. Recovery is defined as the first sample at which every window's relative mean error stays below 5% for 30 consecutive samples; an event that never meets the condition is not silently counted as a success. Rad-DAES recovers 55 of 80 abrupt-step events with a mean recovery time of 4198 s; gated EWMA recovers all 80 at 1734 s but with a much higher false-alarm burden; ungated EWMA and RLS recover all 80 at about 399 s while remaining nearly blind to sources; `static` recovers none. Recovery and detection therefore pull in opposite directions in this matrix, and no method achieves both.

Tuning and fallbacks. Table 4 records the selected development configuration for each method. Four of the nine methods used a fallback, meaning that no candidate met both the background-error

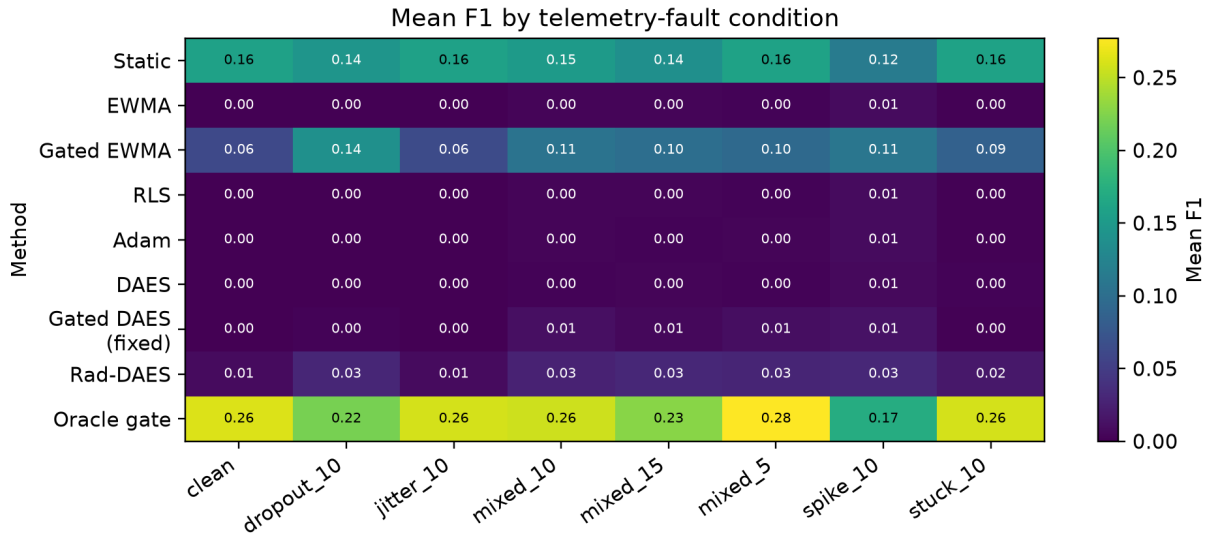


Figure 3: Mean F1 by telemetry-fault condition. Each cell is the mean over the four regimes and five seeds.

and false-alarm constraints on the development scenarios. Rad-DAES selected the largest gain in the grid ($\gamma = 0.5$) with a development FAR of 28.6/h, more than fourteen times the constraint; gated EWMA selected the smallest EWMA weight but still had a development FAR of 99.4/h. The oracle gate selected the smallest gain ($\gamma = 0.05$) and had a development FAR of 12.5/h. The oracle's advantage is therefore not an isolated gate effect: it combines perfect labelling with a ten-fold gentler step, and the two cannot be separated in this matrix. The selected configurations were frozen before the test matrix.

5 Discussion

The experiment was designed to test a trade-off, not to demonstrate that a new method wins. The answer is negative, and three mechanisms explain most of it.

First, the innovation guard confuses drift with a source. A persistent background step produces a sustained score above the gate threshold, so the gate freezes the mean precisely when the mean needs to move. The 90-sample release was intended to break this deadlock, but it requires an uninterrupted run of exceedances; Poisson noise and the oscillating score reset the counter. In the representative trace the counter peaks at 89 and resets. The result is a frozen model, a long alarm burst, and a slow recovery. This is a design failure of the release rule, not a tuning issue: the rule is brittle by construction because it depends on a run length in a noisy sequence.

Second, the adaptive background absorbs source energy. Both the mean and the covariance update on a 50-sample time constant, comparable to the 20–60 s source duration. During the ramp of a source event the score is still below the gate threshold, so the gate passes those samples; the mean then moves toward the source and the covariance learns its outer product, and the score for the peak falls. This is consistent with the threshold-free results: `static`, which updates neither the mean nor the covariance, has ROC AUC 0.728, while the adaptive methods have ROC AUC between 0.558 and 0.691. We did not run a covariance-frozen or mean-frozen ablation, so the two channels are not separated experimentally; the static-versus-adaptive contrast supports the general mechanism, not the share attributable to covariance. It is the most plausible explanation for why

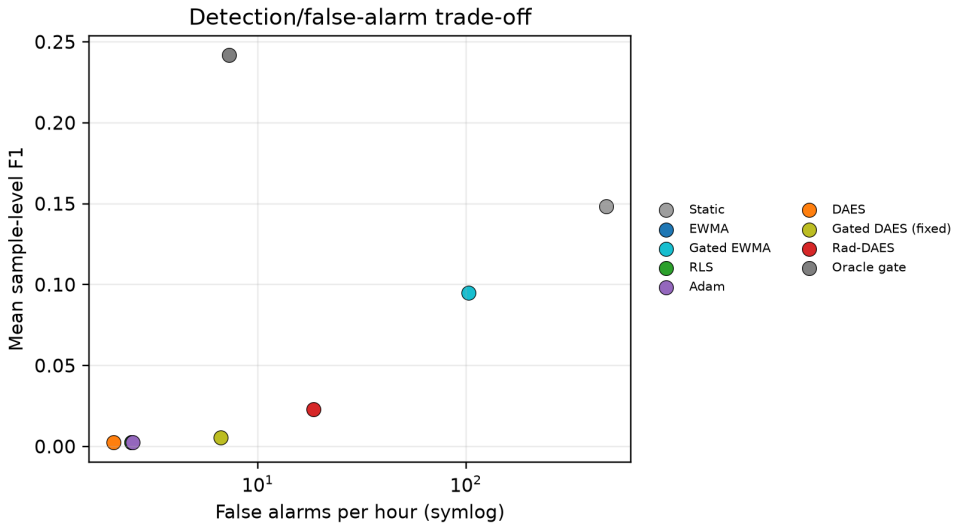


Figure 4: Detection/false-alarm trade-off. The horizontal axis is symlog because static and some gated configurations produce very large drift-induced false-alarm burdens.

the ungated methods detect only about 6% of events despite a nominal 3–8σ source amplitude.

Third, the selection rule was infeasible. The development constraint (FAR $\leq$ 2/h, background error $\leq$ 0.10) could not be met by any gated configuration under abrupt drift. The fallback rule then chose configurations that violated the FAR constraint by one to two orders of magnitude, and those configurations were frozen for test. The four feasible methods were feasible only because they were nearly blind. The protocol therefore did not compare methods at a common, achievable operating point; it compared blind methods against noisy ones. A matched-FAR evaluation, or a prespecified ROC/PR analysis, would have been more informative. The ROC/PR table is reported here as a secondary analysis, but it was not the basis of the selection rule.

The practical implication is that the three decisions should not share one score. A drift detector, a source detector, and a telemetry-integrity check answer different questions; using the same Mahalanobis score for gating and alarming creates the coupling observed here. A two-timescale covariance (fast for the mean, slow or robust for the detector), or a covariance that is frozen during suspected sources, is a concrete next step. None of these is implemented or validated in this draft.

The draft does not establish that Rad-DAES is novel. The literature audit was rapid and non-systematic, and the closest related work on portal data and concept drift is cited only to frame the problem. A novelty claim would require a protocol-driven review and a direct comparison on public data.

6 Limitations and threats to validity

- **No real data.** RADAI requires an account and a large download; PyRIID requires a compatible Python environment and spectra conversion; the JRC reference spectra are not a continuous portal stream. No credentials were used and no real stream was parsed. Gate 1 of the project plan is a partial failure.
- **Synthetic generator.** The regimes, source templates, amplitudes, and fault rates are design choices. They are not fitted to measured background. The Poisson model ignores detector resolution, dead time, and spectrum unfolding. The $\sin^2$ source profile makes the peak narrow,

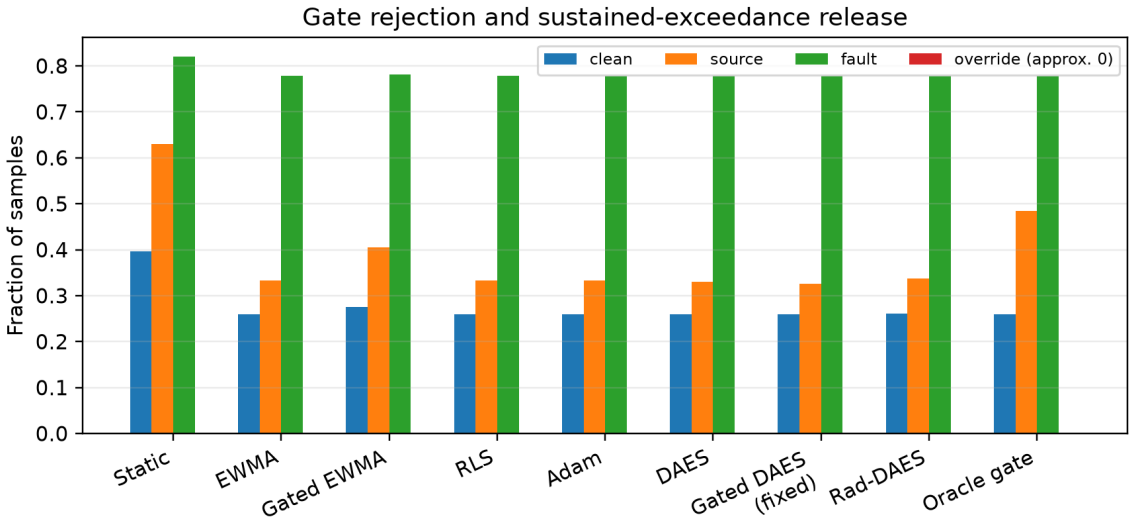


Figure 5: Gate rejection rates for clean, source, and fault samples, and the rate of sustained-exceedance release. A high source-rejection rate is the expected cost of protecting the background; it is also a recall risk.

which penalises sample-level F1; event-level detection is reported to reduce that sensitivity, but it does not eliminate it.

- **Mechanism inferred, not isolated.** The adaptive-background explanation (mean and covariance) is supported by the static-versus-adaptive contrast and by the saved traces, but no covariance-frozen or mean-frozen ablation was run, so the two channels are not separated. The gate-deadlock explanation is directly visible in one representative trace; the 89-versus-90 near miss is illustrative, not a distributional estimate.
- **Small seed budget.** Five seeds per scenario give wide intervals. The intervals describe seed variation only and do not cover scenario uncertainty or model misspecification.
- **No multiplicity correction.** The scenario-level and paired comparisons are descriptive. A claim of statistical significance is not made.
- **Design choices.** The 90-sample (90 s) release, the 0.5–1.5 s timing bound, the 0.25–3.5 Fano bounds, the 0.25 covariance floor, the 50-sample covariance time constant, and the six windows are choices, not validated constants. Sensitivity analyses are future work.
- **Threshold transfer.** Thresholds calibrated on clean stable streams do not transfer to drift regimes: even the oracle gate has 7.3 false alarms/h on test. The 1/h calibration target is therefore not an operational guarantee.
- **DAES provenance.** The DAES equations follow the supplied scalar reference implementation. The matrix update in some generated planning material is not implemented as DAES. The wrapper's use of a Gaussian quasi-likelihood Hessian is an adaptation, not a theorem.
- **No field readiness.** Runtime is offline Python on synthetic arrays. No hardware latency, network delay, calibration uncertainty, or regulatory requirement is modelled.

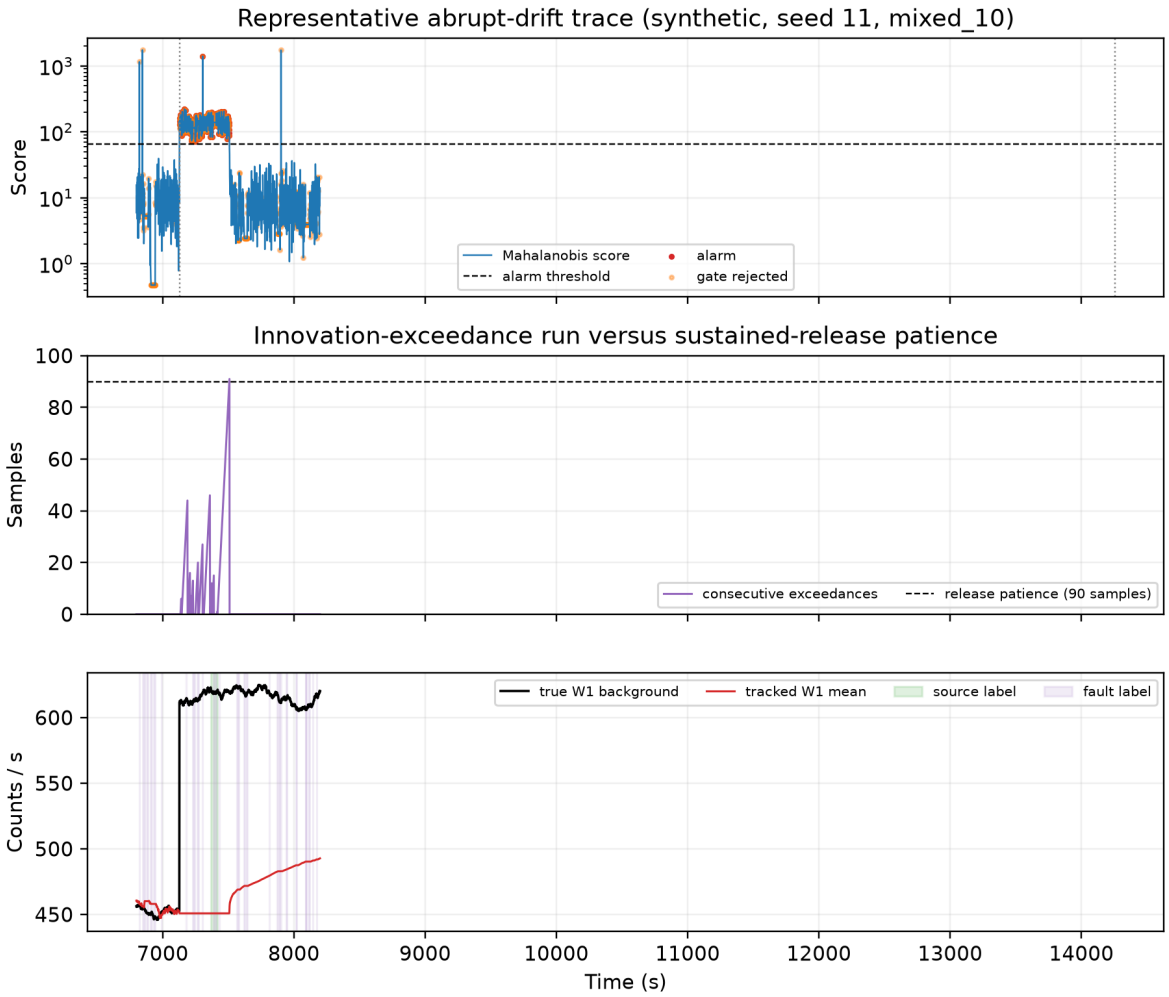


Figure 6: Representative abrupt-drift trace for Rad-DAES (synthetic, seed 11, `mixed_10`). Top: pre-update score, alarms, and gate rejections. Middle: consecutive innovation exceedances versus the 90-sample release patience; the counter peaks at 89 and resets. Bottom: true and tracked W1 background, with source and fault labels.

7 Reproducibility

The repository is self-contained and separate from the unrelated projects that share the workspace. The synthetic generator, gate, DAES controller, background model, detector, runner, tests, scripts, raw CSV files, tables, and figures are included. The main commands are:

```
python -m pip install -e .
python -m pytest -q
python scripts/run_benchmarks.py --mode full --output results/raw
python scripts/aggregate_results.py
python scripts/make_figures.py
python scripts/audit_results.py
```

The runner writes a manifest with Python and package versions, configuration hashes, seed lists, and run counts. Every table and figure is generated from saved CSV or trace files; no result is typed

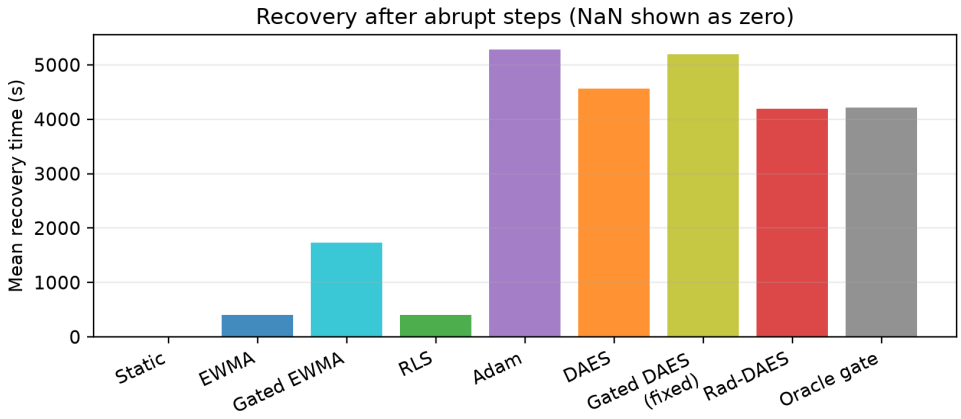


Figure 7: Mean recovery time after abrupt steps. Missing values are shown as zero only for display; they are not treated as recovered in the tables.

into the manuscript by hand. The audit script checks matrix completeness, the frozen selection and thresholds, a determinism spot-check of saved rows, config hashes, and the presence of every table, figure, and macro used here. The original DAES reference file was read from an external snapshot for interface documentation and was not vendored or modified. Its SHA-256 and the archive SHA-256 are recorded in `docs/daes_interface.md`.

The raw run was launched with `--output results` and then moved into `results/raw/` to match the layout read by the aggregation and audit scripts; the only edited field was the trace-path column of `trace_index.csv`. This is documented in `results/raw/README.md`.

8 Conclusion

This draft delivers a small, complete, reproducible synthetic study of guarded drift-aware background tracking, and its result is negative. Rad-DAES does not improve the detection/false-alarm trade-off on these streams: its F1 is 0.023 at 18.508/h, while the four false-alarm-compliant methods are nearly blind and the non-deployable oracle reaches only 0.242. The traces identify two concrete failure modes, gate-induced freezing under drift and adaptive-background contamination during source ramps, and the development selection rule is shown to be infeasible at the prespecified operating point. The value of the draft is the auditable pipeline and the explicit evidence boundary, not a claim that the candidate method is ready for a portal.

References

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Table 3: Paired differences (Rad-DAES minus the named method) over matched scenario/seed pairs. Positive F1 is better; positive FAR and BG are worse. Intervals are descriptive 95% Student-t intervals and are not multiplicity-corrected.

Comparison	Metric	Mean difference	95% CI	Pairs
Rad-DAES minus Static	f1	-0.125	[-0.146, -0.105]	160
Rad-DAES minus Static	far_per_hour	-453.9	[-534.8, -372.9]	160
Rad-DAES minus Static	bg_rel_error	-0.029	[-0.034, -0.024]	160
Rad-DAES minus EWMA	f1	0.020	[0.013, 0.027]	160
Rad-DAES minus EWMA	far_per_hour	16.032	[6.324, 25.741]	160
Rad-DAES minus EWMA	bg_rel_error	0.010	[0.008, 0.012]	160
Rad-DAES minus Gated EWMA	f1	-0.072	[-0.081, -0.063]	160
Rad-DAES minus Gated EWMA	far_per_hour	-84.240	[-122.8, -45.653]	160
Rad-DAES minus Gated EWMA	bg_rel_error	0.008	[0.006, 0.010]	160
Rad-DAES minus RLS	f1	0.020	[0.013, 0.027]	160
Rad-DAES minus RLS	far_per_hour	16.032	[6.324, 25.741]	160
Rad-DAES minus RLS	bg_rel_error	0.010	[0.008, 0.012]	160
Rad-DAES minus Adam	f1	0.020	[0.014, 0.027]	160
Rad-DAES minus Adam	far_per_hour	16.008	[6.291, 25.726]	160
Rad-DAES minus Adam	bg_rel_error	-0.002	[-0.004, -0.000]	160
Rad-DAES minus DAES	f1	0.020	[0.014, 0.027]	160
Rad-DAES minus DAES	far_per_hour	16.478	[6.720, 26.236]	160
Rad-DAES minus DAES	bg_rel_error	-0.012	[-0.017, -0.007]	160
Rad-DAES minus Gated DAES (fixed)	f1	0.017	[0.011, 0.023]	160
Rad-DAES minus Gated DAES (fixed)	far_per_hour	11.860	[2.278, 21.441]	160
Rad-DAES minus Gated DAES (fixed)	bg_rel_error	-0.043	[-0.044, -0.043]	160
Rad-DAES minus Oracle gate	f1	-0.219	[-0.231, -0.207]	160
Rad-DAES minus Oracle gate	far_per_hour	11.217	[1.634, 20.799]	160
Rad-DAES minus Oracle gate	bg_rel_error	0.011	[0.010, 0.012]	160

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Table 4: Selected development configurations and development means. "Fallback" means no candidate satisfied both the background-error and false-alarm constraints.

Method	Selected configuration	Status	Dev F1	Dev BG	Dev FAR/h
Static	`default`	fixed	0.137	0.0644	647.7
EWMA	`ewma_alpha=0.02`	feasible	0.004	0.0136	1.11
Gated EWMA	`ewma_alpha=0.005;gate_quantile=0.999`	fallback	0.078	0.0148	99.36
RLS	`rls_forgetting=0.98`	feasible	0.004	0.0136	1.11
Adam	`adam_learning_rate=0.02`	feasible	0.005	0.0261	1.09
DAES	`daes_gain=0.2`	feasible	0.004	0.0417	1.05
Gated DAES (fixed)	`daes_gain=0.5;gate_quantile=0.995`	fallback	0.011	0.0692	12.42
Rad-DAES	`daes_gain=0.5;gate_quantile=0.999`	fallback	0.036	0.0257	28.64
Oracle gate	`daes_gain=0.05`	fallback	0.228	0.0147	12.48

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A Implementation map

Table 5: Where each component is implemented and tested. Package paths are relative to `src/rad_daes/`; script, test, and report paths are relative to the repository root.

Component	Source	Test
Synthetic streams	`data/synthetic.py`	`tests/test_synthetic.py`
Window definitions	`preprocessing/windows.py`	configuration checks
Background model	`background/model.py`	`tests/test_background.py`
Telemetry gate	`telemetry/gate.py`	`tests/test_gate.py`
DAES controller	`adaptation/daes.py`	`tests/test_daes.py`
Detector	`detection/detector.py`	runner integration
Metrics	`evaluation/metrics.py`	`tests/test_metrics.py`
Runner and calibration	`evaluation/runner.py`	`tests/test_runner.py`
Aggregation	`scripts/aggregate_results.py`	table checks
Audit	`scripts/audit_results.py`	`results/audit_report.md`

B Protocol revisions

The source-placement bug, the sustained-exceedance release, the covariance floor, and the midpoint source profile were fixed on 28 October 2026 before the held-out matrix was run. The revision log

is in `docs/protocol.md`. No held-out result was inspected before these changes. After the frozen test matrix was complete, the threshold-free ROC/PR table and the event-level detection summary were added as secondary analyses; they are computed from columns already saved by the frozen run, and the primary estimand and selection rule were not changed.

C Artifact inventory

Table 6 lists the paths a reader needs in order to inspect or rerun the study. Counts describe the working repository of 28 October 2026 and exclude Python caches and LaTeX build byproducts; the submission mirror differs only in that the cover-letter DOCX sits at the bundle root, so its `paper/` directory has six files.

Table 6: Frozen artifact inventory. File counts exclude `__pycache__` directories.

Path	Contents	Files
`src/rad_daes/`	Generator, gate, DAES controller, model, detector, metrics, runner	16
`configs/`	Data, model, and experiment configuration	3
`tests/`	Unit and integration tests (44 passing)	7
`results/raw/`	Calibration, development, and test CSVs; manifest; 4 traces	13
`results/tables/`	CSV, LaTeX tables, and generated macros	16
`results/figures/`	Seven PNG figures	7
`docs/`	Protocol, decisions, DAES interface, results summary, checklist	8
`paper/`	Manuscript source, PDF, cover letter, bibliography, supplementary note	7

The primary test artifact is `results/raw/test_runs.csv` (1,440 rows). The audit report is regenerated by `scripts/audit_results.py` and records 22 checks. No real radiation data are included, because none were obtained (Section 6).